

NAMING COMPOUNDS

Ionic Compounds

Ionic compounds have a *metal element* combined with a *non-metal element* (or elements).

Use the Periodic table to *identify if a metal element is present*. If there is, then the *substance is ionic*.

Rules for Naming Ionic Compounds

- Name the metal first, then the ion.
- Use the table of common ions (see separate sheet) to help name the compound.

Examples

<u>Formula</u>	<u>Name</u>	<u>Description</u>
MgCO ₃	magnesium carbonate	1xMg, 1xC, 3xO atoms
Zn(OH) ₂	zinc hydroxide	1xZn, 2xO, 2xH atoms
This multiplies everything in the bracket.		
Cu ₃ (PO ₄) ₂	copper phosphate	3xCu, 2xP, 8xO atoms

Covalent Compounds

Covalent compounds have *all non-metal atoms*.

Use the Periodic Table to *identify if a metal is present*. If *not*, then it is *most likely covalent*.

Exception - any substance with the ammonium ion (NH_4^{1+}) in it is ionic.

e.g. ammonium chloride NH_4Cl

Rules For Naming Covalent Compounds

- Use prefixes such as mono-, di-, tri-, etc.

i.e.	mono -	one	penta -	five
	di -	two	hexa -	six
	tri -	three	hepta -	seven
	tetra -	four	octa -	eight

Examples

<u>Formula</u>	<u>Name</u>	<u>Description</u>
SO ₃	sulphur trioxide	1xS, 3xO atoms
CO ₂	carbon dioxide	1XC, 2xO atoms
N ₂ O ₄	dinitrogen tetroxide	2xN, 4xO atoms
SF ₆	sulphur hexafluoride	1XS, 6xF atoms